computer science & engineering

Risk-Adjusted Hallucination Detection

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Motivation

Large Language Models (LLMs) are increasingly used in critical decision-making, but they often produce fluent yet incorrect responses.

- **High-Stakes Risk**: In domains such as healthcare, law and finance, small errors can lead to significant harm
- **"Confident" Mistakes**: LLMs often generate incorrect responses with high confidence, making them difficult for users to detect without assistance.
- **Signal Limitation:** Existing detection methods often rely on only one signal, which cannot fully capture hallucination behaviour.

Literature Survey

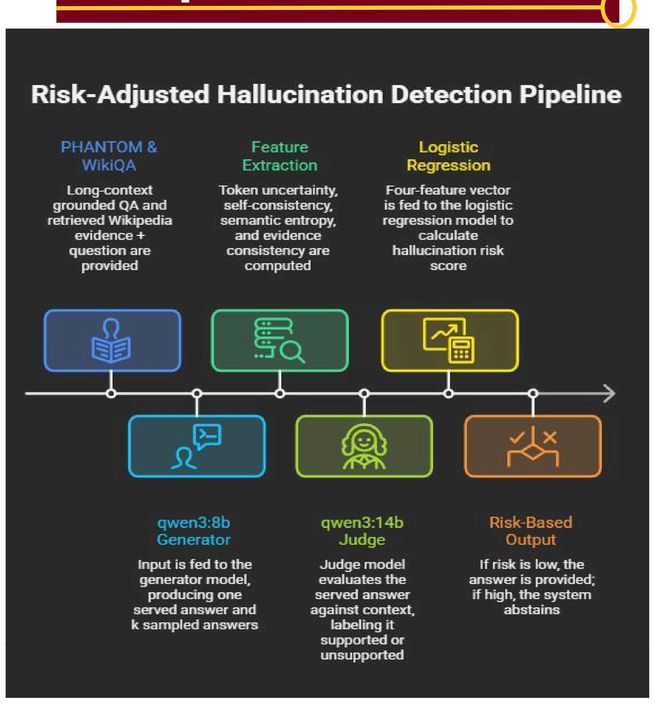
The project builds upon several established research directions:

- **Uncertainty & Consistency**: Methods like SelfCheckGPT (Manakul et al., 2023) and semantic entropy (Farquhar et al., 2024) measure disagreement and meaning-level uncertainty.
- **Calibration**: Techniques such as temperature scaling reduce model overconfidence and improve reliability (Guo et al., 2017).
- **Selective Abstention**: Models can abstain when uncertain, trading coverage for accuracy (Geifman and El-Yaniv, 2017)
- **Evidence Groundedness**: NLI and LLM-based evaluator check if responses are supported by evidence (Bowman et al., 2015), (Zheng et al., 2023).

Problem Definition

- **Task:** Detect when AI-generated answers lack supporting evidence
- **Challenge**: Hallucinations vary across evidence regimes and tasks, with no obvious detection signal
- **Gap**: Single-signal methods miss many failure modes; uncertainty misses confident errors, and evidence-based methods depend on data quality
- **Evaluation Objective**: Testing if a risk-aware detector works well within a dataset and if it generalizes across PHANTOM and WikiQA

Proposed Ideas

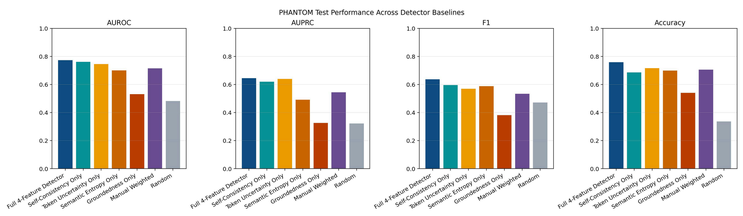


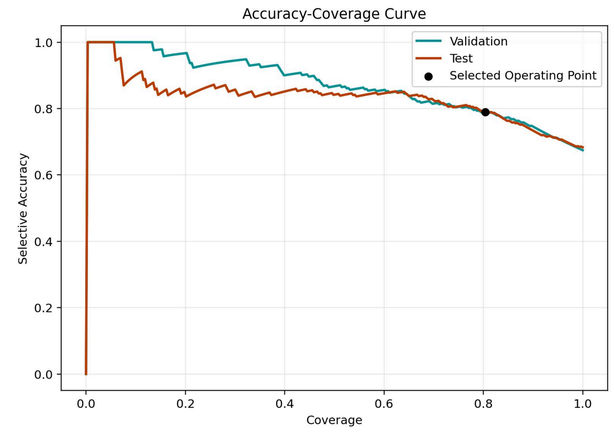
Contribution

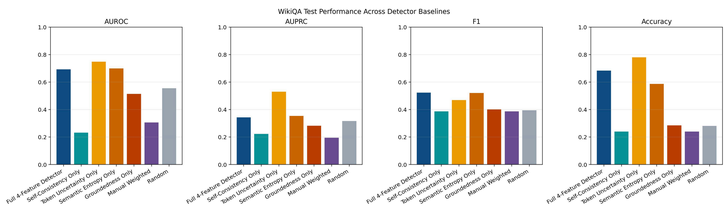
- Built a 4-feature risk-adjusted hallucination detector that uses token-level entropy, self-consistency disagreement, semantic entropy, and evidence groundedness.
- Evaluated the hallucination detector on 2 evidence regimes: PHANTOM and WikiQA.
- Applied calibration and selective abstention to avoid high-risk answers.
- Cross-dataset transfer tested from PHANTOM to WikiQA and vice versa to quantify detector generalization under data distribution shift.

Experimental results

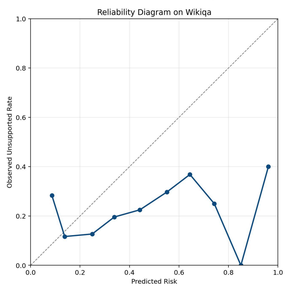
In-domain results for PHANTOM and WikiQA

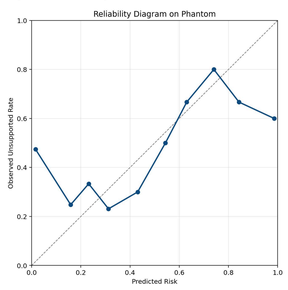






Cross-domain transfer results (PHANTOM → WikiQA & WikiQA → PHANTOM





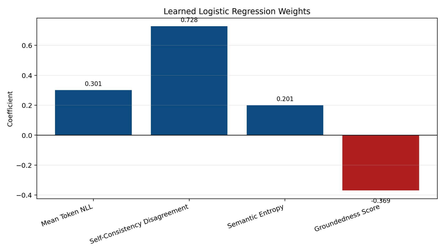
Limitation and discussion

- **Dataset dependence**: Performance on PHANTOM is strong but weak on WikiQA, showing sensitivity to evidence quality and regime.
- **Feature instability**: Uncertainty signals (token and semantic entropy, self-consistency) vary across datasets, limiting generalization.
- **Poor cross-domain transfer**: Detectors do not transfer well between PHANTOM and WikiQA in either of the directions, indicating weak generalization.
- **Calibration sensitivity**: Best calibration differs by dataset (Platt for PHANTOM vs isotonic for WikiQA), preventing a universal approach.
- **Weak stand-alone grounding signal**: Evidence consistency as a stand-alone signal is insufficient but helps in hallucination detection when combined with other features.
- **Limited abstention gains**: Abstention clearly helps on PHANTOM data but offers little improvement on noisier WikiQA.
- **Retrieval dependency**: Performance on WikiQA is strongly affected by retrieval quality.
- **Key takeaway**: Better cross-domain robustness needs improvements in feature stability, retrieval quality, and lightweight adaptation methods.

Main findings

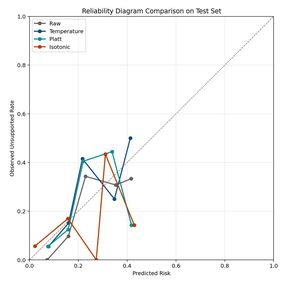
PHANTOM

- Strongest in-domain hallucination detector
- AUROC: 0.7730; AUPRC: 0.6451; F1: 0.6368
- The full 4-feature detector outperforms single-feature baselines
- Learned weights in the detector are intuitive
  - Uncertainty-based features increase risk of being a hallucination
  - Evidence groundedness decreases risk of being a hallucination
- Abstention leads to an improvement in retained-answer quality from 0.6832 to 0.7863



WikiQA

- Harder evidence regime
- AUROC: 0.6927; AUPRC: 0.3434; F1: 0.5231
- The full 4-feature detector contains useful signal, but the gap from the single-feature baselines is smaller in comparison to PHANTOM
- Feature behaviour is not as stable as PHANTOM, so the learned weights in the detector are not entirely intuitive
- Isotonic regression is better suited at calibrating this dataset compared to Platt scaling which was used for PHANTOM, indicating that calibration choice is dataset-dependent



Cross-domain transfer

- Generalization is weak in both directions
- PHANTOM → WikiQA: AUROC 0.3996 and AUPRC: 0.2020
- WikiQA → PHANTOM: AUROC 0.4294 and AUPRC: 0.2943
- When moved to a new dataset, the detector's risk scores do not match the true unsupported examples rate well enough
- The primary bottleneck is feature instability
  - Evidence groundedness remains useful
  - However, uncertainty-based features vary more across datasets

Plan for the final report

- Present a unified and detailed description of the entire pipeline across PHANTOM and WikiQA datasets
- Provide a summary of the most important results with all tables and figures included
- Perform a comparison of in-domain and cross-domain performance and interpret the reason behind transfer failing
- Provide qualitative examples and failure cases from PHANTOM and WikiQA
- Discuss feature instability, potential improvements for better transfer, and calibration shift

Citations

- Potsawee Manakul, Adian Liusie, and Mark Gales. 2023. Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models. In Proceedings of the 2023 conference on empirical methods in natural language processing, pages 9004–9017.
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